UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

# Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 011 (Legendre) · Legendre's and Cauchy's linear differential equations · compiled 17 October 2026

| | |
|---|---|
| ESA ★★★ | Asked repeatedly in past ESA papers |
| ESA ★★ | Asked in past ESA papers or computed in class-work problems |
| ESA ★ | Conceptual: taught in class, not examined so far |
| ADDED FOR CLARITY | Standard context that is not in the class material |
| TRANSCRIPT ONLY | Said in the lecture but not on any slide |
| VIDEO ONLY | Seen only in a frame of the class video |
| NOT ON CURRENT SLIDES | From older material or past papers, not this year's slides |
| S011L·8 | Unit 2 deck 011 "Legendre's differential equation" (SESSION 9), slide 8 (rendered .pptx) |
| T011L 00:05:20 | AV-summary transcript "Legendre differential equation" (21.9 min, listed first), segment start |
| T011Lb 00:12:26 | AV-summary transcript "Legendre differential equations_Continued" (23 min, second), segment start |
| CW U2 P30 | Unit 1 & 2 class-work sheet, Unit 2 problem 30 |
| ESA Jul-23 3a | Past paper: UE22MA141A July 2023, question 3.a (Jan-24 s1...s4 = UE23MA141A Jan 2024 sets 1–4) |

## Contents


1. 1 · Linear equations with variable coefficients: Legendre's and Cauchy's forms
2. 2 · The substitution $ax + b = e^z$ and the operator results
3. 3 · Legendre problems: the slide example, CW P29–P30 and five ESA questions
4. 4 · Cauchy problems: lecture examples, CW P31–P33 and ESA Jan-24 s3 3d
5. Past-paper questions


# 1 Linear equations with variable coefficients: Legendre's and Cauchy's forms

### Definitions

**Legendre's linear differential equation.** An equation of the form $k_0(ax+b)^n \frac{d^n y}{dx^n} + k_1(ax+b)^{n-1}\frac{d^{n-1}y}{dx^{n-1}} + k_2(ax+b)^{n-2}\frac{d^{n-2}y}{dx^{n-2}} + \cdots + k_n y = X$, where the $k_i$ are constants ($k_0 \neq 0$) and $X$ is a function of $x$; it is of order $n$. S011L·4 · S011L·5 · T011L 00:01:38 · T011L 00:02:05

**Cauchy's linear differential equation.** An equation of the form $k_0x^n\frac{d^ny}{dx^n}+k_1x^{n-1}\frac{d^{n-1}y}{dx^{n-1}}+k_2x^{n-2}\frac{d^{n-2}y}{dx^{n-2}}+\cdots+k_ny=X$, $X$ a function of $x$: Legendre's equation with $a=1$, $b=0$ (also called the Cauchy–Euler or Euler equation). S011L·4 · T011L 00:02:16 · T011L 00:16:51 · T011L 00:17:52

**Variable coefficients.** The coefficients of the derivatives are functions of the independent variable $x$ (here $(ax+b)^k$ or $x^k$), not constants. T011L 00:03:10 · S011L·4

## Statements in the class material

> *"We consider two forms of differential equations with variable coefficients which can be reduced to linear differential equations with constant coefficients by suitable substitutions."* S011L·4

## Key points

1. Recognise the form: in every term the degree of the coefficient equals the order of the derivative ($(ax+b)^2$ with $y''$, $(ax+b)$ with $y'$, $(ax+b)^0$ with $y$). ESA ★★★ T011L 00:04:06 · T011L 00:04:46 · T011L 00:15:00 · T011Lb 00:01:31
2. Cauchy's equation is Legendre's with $a=1$, $b=0$; the same method works with $x=e^z$. ESA ★★★ S011L·4 · T011L 00:02:16 · T011L 00:17:52 · T011Lb 00:03:24
3. An equation that is not in the form can sometimes be brought to it: $xy''''+10y'''=0$ times $x^3$ is Cauchy's equation $x^4y''''+10x^3y'''=0$; $y''+\frac{1}{x}y'=\ldots$ times $x^2$ (CW P32). ESA ★★ transcript only T011Lb 00:18:25 · T011Lb 00:18:37 · CW U2 P32
4. Cauchy's equation is the syllabus "self-learning component", but it is asked (ESA Jan-24 s3 3d = s4 3c) and has three CW problems. ESA ★★★ not on current slides CW U2 P31 · ESA Jan-24 s3 3d · ESA Jan-24 s4 3c

## Exam traps

***Exam trap.*** Slides 011L·4–5 print $k_2(ax+b)^{n-2}\frac{d^{n-1}y}{dx^{n-1}}$ (and $k_2x^{n-2}\frac{d^{n-1}y}{dx^{n-1}}$): the derivative is $\frac{d^{n-2}y}{dx^{n-2}}$ — degree equals order. S011L·4 · S011L·5 · T011L 00:04:06

# 2 The substitution $ax + b = e^z$ and the operator results

### Definitions

**The substitution.** $ax + b = e^z$, i.e. $z = \log(ax + b)$ and $x = \frac{e^z - b}{a}$; for Cauchy $x = e^z$, $z = \log x$. It changes the independent variable from $x$ to $z$. S011L·5 · T011L 00:05:20 · T011Lb 00:01:49 · T011L 00:18:30

**$D$ after the substitution.** $D = \frac{d}{dz}$, the differential operator with respect to $z$ (some books write $\theta$ ); before the substitution $D$ meant $\frac{d}{dx}$. S011L·6 · T011L 00:07:44 · T011Lb 00:02:04

### Statements in the class material

> *"Such equations can be reduced to linear differential equations with constant coefficients by the substitution* $ax + b = e^z$ *or* $z = \log(ax + b)$, $\frac{dz}{dx} = \frac{a}{ax+b}$*."* S011L·5

> *"Using these results in Legendre's Differential Equation, we get Linear Differential Equation with constant coefficients which can be solved by the methods discussed before."* S011L·6

> *"Consider Second order Legendre's Linear Differential Equation* $(ax + b)^2 \frac{d^2y}{dx^2} + k_1(ax + b)\frac{dy}{dx} + k_2y = X$*. By the substitution* $z = \log(ax + b)$*, the above equation reduces to* $(a^2D(D - 1) + k_1aD + k_2)y = f(z)$*. It is a second order linear non homogeneous differential equation with constant coefficients which can be solved."* S011L·7

### Key points

1. Four steps (F2.1): substitute $z = \log(ax + b)$ → write the equation and the RHS in $z$ → CF and PI in $z$ → back to $x$. ESA ★★★ S011L·7 · S011L·8 · T011L 00:13:26 · T011Lb 00:12:12
2. Remember $(ax + b)^2y'' = a^2D(D - 1)y$, not $a^2D^2y$: the $-D$ comes from the product rule. ESA ★★★ S011L·6 · T011L 00:11:45 · T011L 00:12:34
3. The factor $a$ matters: $(2x + 3)y' = 2Dy$ and $(2x + 3)^2y'' = 4D(D - 1)y$. ESA ★★★ S011L·8 · T011Lb 00:13:23 · T011Lb 00:13:37
4. The right-hand side must also be written in $z$ (use $x = \frac{e^z - b}{a}$); the final answer must be in $x$ again. ESA ★★★ T011L 00:15:57 · T011L 00:21:15 · T011Lb 00:09:29 · T011Lb 00:17:39
5. After the substitution $D$ means $\frac{d}{dz}$, and a failure case gives $z\,e^{kz}$, not $x\,e^{kx}$. ESA ★★★ T011L 00:07:44 · T011L 00:12:03 · T011Lb 00:11:07

### Why?

**Why does $ax + b = e^z$ remove the variable coefficients?**
Since $\frac{dz}{dx} = \frac{a}{ax+b}$, each $x$-derivative brings a factor $\frac{1}{ax+b}$; the coefficient $(ax + b)^k$ in front of the $k$-th derivative cancels them exactly, leaving constants times $z$-derivatives. That is why the degree must equal the order.

ADDED FOR CLARITY T011L 00:06:28 · T011L 00:13:26

**Where does the $D-1$ in $a^2D(D-1)$ come from?**
$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{a}{ax+b}\frac{dy}{dz}\right) = -\frac{a^2}{(ax+b)^2}\frac{dy}{dz} + \frac{a^2}{(ax+b)^2}\frac{d^2y}{dz^2}$: the first term (derivative of $\frac{a}{ax+b}$) gives $-D$.
T011L 00:09:33 · T011L 00:10:52 · T011L 00:11:45

**Why does $e^{mz}$ become a power of $ax+b$?**
$e^{mz} = (e^z)^m = (ax+b)^m$. So Legendre and Cauchy solutions are powers (and logs), not exponentials, in $x$.
S011L·9 · T011Lb 00:15:21

## Exam traps

***Exam trap.*** Forgetting the powers of $a$: $(2x+1)^2y'' = 4D(D-1)y$, not $D(D-1)y$. S011L·8 · ESA Jul-23 3a

***Exam trap.*** Leaving the right-hand side in $x$ (e.g. solving $(D^2-2D-3)y = 6x$ in $z$) is wrong: write $x = \frac{e^z-3}{2}$ first. S011L·8 · T011L 00:15:57

***Exam trap.*** Answers left in $z$ lose marks: replace $e^z$ by $ax+b$ and $z$ by $\log(ax+b)$. T011Lb 00:09:29 · T011Lb 00:17:39

# 3 Legendre problems: the slide example, CW P29–P30 and five ESA questions

## Key points

1. Polynomial right-hand sides give $e^{kz}$ terms plus a constant: ESA Jan-24 s1 3c ($6x+5 \to 3e^z+2$), s2 3c ($3x+4 \to 3e^z-2$), Jun-24 3b ($8x^2-2x+3 \to 2e^{2z}+3e^z+4$), CW P30 ($\to \frac{e^{2z}-1}{3}$). ESA ★★★ ESA Jan-24 s1 3c · ESA Jan-24 s2 3c · ESA Jun-24 3b · CW U2 P30 · S011L·8
2. Check every $e^{kz}$ against the AE roots: the failure case appears in Jul-23 3a ($(D-2)^2$, twice), Jan-24 s2 3c ($(D-1)^2$, twice), Jun-24 3b ($e^z$, root 1), CW P29 ($\sin z$, roots $\pm i$) and CW P30 ($e^{2z}$, root 2). ESA ★★★ ESA Jul-23 3a · ESA Jan-24 s2 3c · ESA Jun-24 3b · CW U2 P29 · CW U2 P30
3. A repeated root $m$ gives $(ax+b)^m[c_1+c_2\log(ax+b)]$; complex roots $\pm i\beta$ give $c_1\cos(\beta\log(ax+b))+c_2\sin(\beta\log(ax+b))$. ESA ★★★ ESA Jul-23 3a · CW U2 P29 · ESA Dec-23 3b

## Why?

**Why is a constant on the right handled as $e^{0z}$?**
$c=ce^{0\cdot z}$, so Type 1 gives $\frac{c}{f(0)}$ when $f(0)\neq 0$ (slide 011L·10: $-\frac{9}{4}\cdot\frac{1}{-3}$).
S011L·10 · T011Lb 00:16:27

## Exam traps

***Exam trap.*** CW P29's printed answer "$-\log(1+x)-\cos\log(1+x)$" should be the product $-\log(1+x)\cos[\log(1+x)]$. CW U2 P29

***Exam trap.*** CW P30's printed answer repeats $(2+3x)^2$; the second CF term is $(2+3x)^{-2}$. CW U2 P30

***Exam trap.*** ESA Dec-23 3b ($\sin(2\log(1+x))$, no failure) and CW P29 ($2\sin\log(1+x)$, failure) look alike but differ. ESA Dec-23 3b · CW U2 P29

# 4 Cauchy problems: lecture examples, CW P31–P33 and ESA Jan-24 s3 3d

## Key points

1. $x = e^z$: $x^m \to e^{mz}$, $\frac{1}{x} \to e^{-z}$, $\log x \to z$; so $x^2 \log x$ is $ze^{2z}$ (Type 4, class 006). ESA ★★★ T011Lb 00:06:23 · CW U2 P31 · CW U2 P33
2. Roots $-1$, $1 \pm i$ in $z$ give $\frac{c_1}{x} + x[c_2 \cos(\log x) + c_3 \sin(\log x)]$. ESA ★★ TRANSCRIPT ONLY T011Lb 00:08:47 · T011Lb 00:09:11
3. ESA Jan-24 s3 3d: $e^{4z} \sin(e^z)$ is not a standard form; factor $(D-2)(D-3)$ and integrate twice, or use variation of parameters in $x$; $y = c_1x^2 + c_2x^3 - x^2 \sin x$. ESA ★★★ ADDED FOR CLARITY ESA Jan-24 s3 3d · ESA Jan-24 s4 3c

## Exam traps

***Exam trap.*** CW P31 prints $-\frac{x^3}{3}(\log x + \frac{2}{3})$ and CW P32 prints $-2(\log x)^2$; the correct PIs are $-\frac{x^2}{3}(\log x + \frac{2}{3})$ and $2(\log x)^3$. CW U2 P31 · CW U2 P32

***Exam trap.*** CW P32 is not in Cauchy form as printed: multiply by $x^2$ first ($x^2y'' + xy' = 12 \log x$). CW U2 P32

# ★ Past-paper questions

Eight ESA papers checked: UE22MA141A Dec 2023, Jul 2023, May 2024; UE23MA141A Jan 2024 sets 1–4, June 2024 (19 distinct Unit 2 questions, 24 occurrences). Legendre/Cauchy is the most-asked Unit 2 topic: 6 distinct, 8 occurrences, in all 8 papers, 6–7 marks each. Legendre: Dec-23 3b, Jul-23 3a = May-24 3a, Jan-24 s1 3c, s2 3c, Jun-24 3b; Cauchy: Jan-24 s3 3d = s4 3c. Substitution, operator results and RHS conversion ★★★; polynomial RHS (4×) and PI failure cases (4 of 6) ★★★; $\sin(k\log(ax+b))$ (1× + CW P29) ★★; general $n$-th order result and order-3/4 Cauchy (lecture only) ★★; multiplying $xy'''' + 10y''' = 0$ into Cauchy form ★.

### 3 · Legendre problems: the slide example, CW P29–P30 and five ESA questions

**ESA Dec 2023** (6 marks) UE22MA141A Q3.b · solved in W3.3

Solve $(1+x)^2\frac{d^2y}{dx^2} + (1+x)\frac{dy}{dx} + y = \sin 2(\log(1+x))$.

**ESA Jul 2023** (7 marks) UE22MA141A Q3.a · solved in W3.5

Solve $(2x+1)^2\frac{d^2y}{dx^2} - 6(2x+1)\frac{dy}{dx} + 16y = 8(2x+1)^2$.

**ESA May 2024** (7 marks) UE22MA141A Q3.a · solved in W3.5

Solve $(2x+1)^2\frac{d^2y}{dx^2} - 6(2x+1)\frac{dy}{dx} + 16y = 8(2x+1)^2$ (repeat of Jul-23 3a).

**ESA Jan 2024 set 1** (6 marks) UE23MA141A Q3.c · solved in W3.6

Solve $(2x+1)^2y'' - 2(2x+1)y' - 12y = 6x+5$.

**ESA Jan 2024 set 2** (6 marks) UE23MA141A Q3.c · solved in W3.7

Solve $(x+2)^2y'' - (x+2)y' + y = 3x+4$.

**ESA Jun 2024** (7 marks) UE23MA141A Q3.b · solved in W3.8

Solve $(2x-1)^2\frac{d^2y}{dx^2} + (2x-1)\frac{dy}{dx} - 2y = 8x^2 - 2x + 3$.

### 4 · Cauchy problems: lecture examples, CW P31–P33 and ESA Jan-24 s3 3d

**ESA Jan 2024 set 3** (6 marks) UE23MA141A Q3.d · solved in W4.6

Solve $x^2\frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + 6y = x^4\sin x$.

ESA Jan 2024 set 4 · 6 marks · UE23MA141A Q3.c · solved in W4.6

Solve $x^2\frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + 6y = x^4 \sin x$ (repeat of set 3 Q3.d).